

DRIVER EARNINGS NAVIGATOR

Smart Relocation. Better Earnings.

Machine learning-powered recommendations to help NYC Uber drivers wait in the right zones.



THE PROBLEM

During slower periods, drivers may wait in weak-demand zones or relocate without knowing whether the travel time is worth it.



Unpredictable Demand



Wasted Time & Travel



Guesswork, Not Data

OUR SOLUTION

Driver Earnings Navigator recommends the best nearby TLC taxi zone to wait in based on historical Uber trip patterns, estimated travel time, and adjusted earning exposure.



Data-Driven Recommendations



Travel Time Considered



Adjusted Earning Exposure



Visual Map & Clear Explanations

WHO WE HELP



Full-Time Drivers
Work long shifts and need better decisions during slow periods.



Part-Time / Newer Drivers
Need guidance and spatial awareness to find stronger opportunity.

KEY FEATURES



Current Zone or Location
Choose your current TLC zone or use your device location instantly.



Custom Day & Time
Compare current conditions or plan a future shift with what-if scenarios.



Smart Recommendation
Get the best zone, travel time, adjusted earning exposure, and top alternatives.



Zone Map Visualization
See current, recommended, and alternative zones highlighted on the NYC taxi-zone map.



AI Copilot (Gemini)
Ask questions, run scenarios, or use voice to get answers naturally.

Why It Matters

A few smart moves during slow periods can lead to more rides, higher earnings, and less unproductive waiting.



APP DEMO

1. PLAN YOUR MOVE

Best Zone To Wait

Current zone: Midwood (165)

Current location: Use current location

Recommend zone

Howard Beach

Howard Beach is your current zone, so staying here requires no travel time. This reflects overall market activity across all drivers, not individual driver earnings.

Travel time: 0 minutes

Net change after relocating: \$0.00/hr

Adjusted earning exposure: \$1597.11

Inputs used: Howard Beach, Wednesday, 3:00 PM

See more +

2. RECOMMENDATION RESULT

Best Zone To Wait

Using current device location

Recommend zone

East New York

East New York is a strong relocation option. It requires about 11.5 minutes of travel time, run moving there increases your exposure to available ride earnings by \$1055.16 per hour compared to staying in your current zone.

Travel time: 11.5 minutes

Net change after relocating: +\$1055.16/hr

Adjusted earning exposure: \$1597.45

Current zone: Howard Beach,

Inputs used: Howard Beach, Wednesday, 3:00 PM

See more +

3. ZONE MAP

Zone Map

Zone map visualization showing current, recommended, and top alternative zones.

4. AI COPILOT CHAT

VOICE COPILOT

Talk To The Relocator

Mic ready

Copilot

I can help you figure out where to relocate, explain why a zone was recommended, or run what-if scenarios with a different zone, day, or time.

You

What if I start driving at 5 PM in East New York?

Copilot

If you start at 5 PM in East New York, I recommend Howard Beach. You could increase your exposure by approximately \$1597.45/hr after accounting for the 11.5-minute travel time.

Ask a follow-up or run a scenario...

Send **Start mic** **Use location**

5. TOP ALTERNATIVES (HOWARD BEACH EXAMPLE)

Top options right now:

- East New York**
11.5 min +\$1055.16/hr
- South Ozone Park**
12.8 min +\$468.00/hr
- Howard Beach**
0.0 min +\$0.00/hr

Inputs used

Howard Beach, Wednesday, 3:00 PM

MODEL PERFORMANCE

Model Summary

- Model: LightGBM Regressor
- Target: net_gain
- Training Rows: 3,185,960
- Direct Features: PUlocationID, DOlocationID, hour_bucket, day_of_week_numeric, average_PU_to_DO_time

Model Performance (Test Set)

MAE	317.4087
RMSE	460.4430
R ² Score	0.8854
Average NDCG	0.9749

Historical Impact (Randomized Tests)

Across 42,367 random historical contexts, following the model's recommended zone showed higher average earning exposure than staying put.

Average gain per test: +\$752.36/hr

(Represents historical market exposure, not guaranteed individual pay.)



DATA & ML PIPELINE



SYSTEM ARCHITECTURE



TESTING & USER FEEDBACK

Manual Testing (Summary)

- Backend health, zones load, map render
- Recommendations match model output
- Device location works (fetchnor delay)
- Custom day/hour scenarios work
- Clear error messages for failures

User Feedback Highlights

- Removed offered-trip destination prediction (risk to ratings & perks was not worth it).
- Kept explanation text – drivers said it aligns with real-world intuition.
- Added Gemini copilot (chat/voice)

Built for NYC Uber Drivers.
Powered by Data. Guided by Intelligence.

Tech Stack



Smarter Moves.
Better Earnings.

By Bodrul Jalal